CHARITH NUKALA

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PROFESSIONAL SUMMARY

- Data Analyst with 4+ years of experience delivering analytics solutions across finance and healthcare using SQL, Python, Power BI, and Tableau to drive measurable business outcomes at enterprise scale.
- Expertise in data modeling, ETL development, and cloud data warehousing (AWS, Azure, GCP) with hands-on Snowflake and Databricks experience supporting large-scale distributed production environments.
- Skilled in statistical analysis, A/B testing, and predictive modeling to validate business strategies, reduce operational inefficiencies, and support data-driven planning across multi-million-dollar functions.
- Proven Agile/Scrum collaborator with a track record of translating complex business requirements into scalable analytics solutions while mentoring peers and championing data-driven practices across teams.

SKILLS

Data Analysis & Visualization: SQL, Python (Pandas, NumPy), Tableau, Power BI, Excel (Power Query, PivotTables), KPI Dashboards, Data Storytelling, Business Intelligence, Looker

Data Engineering & ETL: ETL Pipelines, Apache Airflow, Spark, Databricks, Snowflake, Hadoop, Parquet, JSON

Cloud Platforms: AWS (S3, EC2, Redshift, Glue, Lambda, Athena, CloudWatch, RDS), Azure, GCP (BigQuery, Compute Engine), Kubernetes

Statistical Analysis & Modeling: A/B Testing, Hypothesis Testing, Regression Analysis, Time Series Analysis, Predictive Modeling, scikit-learn, SAS

Tools & Technologies: Jupyter Notebook, Git/GitHub, SQL Workbench, Postman, VS Code, Salesforce, CI/CD, JIRA, Confluence

Programming: Python, SQL, R, JavaScript (ES6+), Shell Scripting, MATLAB, Java, HTML, CSS, TypeScript, React.js

PROFESSIONAL EXPERIENCE

State Farm — Data Analyst Oct 2023 – Present | Irving, TX

- Designed scalable insurance data models in Snowflake and Databricks using advanced SQL (CTEs, window functions, partitioning), improving operational efficiency by **22%** across multi-million-dollar enterprise workflows.
- Automated multi-source reporting workflows using Python (Pandas, NumPy) and Excel (VBA) integrated with AWS S3 pipelines, reducing manual effort by **12+ hours/week** and saving **~$75K annually** across business functions.
- Conducted exploratory analysis and A/B testing (regression, hypothesis testing) to identify inefficiencies, reducing operational costs by **18%** and generating over **$200K in annual savings** across enterprise business units.
- Built dynamic KPI dashboards in Power BI and Tableau integrated with Snowflake and AWS, streamlining ETL data flows and improving executive reporting efficiency and decision-making speed by **30%.**
- Developed predictive forecasting models with Python and scikit-learn using feature engineering and cross-validation, increasing prediction accuracy by **25%** for business units managing budgets exceeding $5M.
- Optimized complex SQL queries (joins, window functions, partitioning) in Snowflake and Databricks production environments, improving pipeline performance and reducing average query execution time by **35%.**
- Implemented data quality validation and anomaly detection frameworks aligned with CI/CD practices, improving data integrity by **20%** across critical enterprise reporting systems and governance workflows.
- Led Agile/Scrum sprint planning and requirements sessions with product, finance, and operations teams via Jira, accelerating cross-functional decision-making speed by **25%** and ensuring on-time delivery.

Tata Consultancy Services (TCS) — Data Analyst Jul 2021 – Aug 2023 | Hyderabad, India

- Analyzed and transformed large-scale healthcare datasets using advanced SQL (CTEs, window functions) across AWS (S3, Redshift, Glue), Azure, and GCP (BigQuery), improving reporting accuracy by **25%.**
- Designed interactive BI dashboards in Power BI, Tableau, and SAS with Parquet-optimized data integration, surfacing actionable clinical and operational insights that improved business efficiency by **20%.**

- Built automated ETL pipelines using Python, Apache Airflow, and Spark, reducing manual processing effort by **30%** and generating approximately **$60K in annual operational savings** through optimized resource utilization.
- Applied statistical forecasting and cost analysis models across cloud data warehouses to identify inefficiencies, contributing to **$150K in annual cost reductions** across key healthcare operations and planning functions.
- Optimized SQL reporting logic and standardized deployments with Git/GitHub version control, collaborating with engineering teams to reduce reporting turnaround time by **30%** across analytics pipelines.
- Designed controlled A/B experiments to evaluate clinical intervention strategies, uncovering statistically significant trends that drove measurable improvements in healthcare service delivery and patient outcomes.
- Maintained HIPAA-aligned data governance and regulatory compliance across cloud environments, establishing access controls, audit logging, and data lineage practices to protect sensitive patient information.
- Collaborated with analysts, engineers, and clinical stakeholders in Agile/Jira environments, contributing to sprint planning and backlog prioritization that improved decision-making velocity by **20%**.

PROJECTS

Customer Churn Prediction & Analytics Platform | Python, SQL, AWS, Power BI, scikit-learn

- Built an end-to-end churn prediction pipeline using Python (XGBoost, scikit-learn, Pandas) and PostgreSQL, engineering 40+ features from 500K+ customer records to achieve **89% model accuracy** with an AUC-ROC of 0.93 across cross-validated test sets.
- Automated ETL workflows on AWS (S3, Lambda, Glue) to ingest and transform customer behavioral data on a scheduled basis, reducing pipeline processing time by **40%** while ensuring consistent data quality for downstream modeling.
- Designed a real-time Power BI dashboard tracking 12+ KPIs (churn probability, CLV, retention rate), enabling stakeholders to identify at-risk segments and target retention strategies projecting a **17% reduction in churn**.
- Applied SHAP analysis for model interpretability, quantifying feature-level contributions and surfacing the top 5 churn drivers to deliver clear, prioritized recommendations to product and customer success leadership.

EDUCATION

Master of Science in Information Systems GPA: 3.44
Cleveland State University | Cleveland, OH

CERTIFICATIONS

- Python for Data Analysis (NumPy, Pandas)
- Generative AI & Agentic AI Fundamentals
- Microsoft Excel & Power BI for Data Analysis and Visualization